

# Named Entity Recognition for Low-Resource Languages

## Experiments, Evaluation and Django Integration

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# Problem Statement

Named Entity Recognition (NER) is a fundamental task in Natural Language Processing used to identify entities such as people, locations, and organizations from raw text.

However, in low-resource languages, building accurate NER systems is challenging due to:

- Lack of sufficient high-quality annotated datasets
- Presence of noisy and inconsistent labels in available data
- Difficulty in generalizing models to real-world sentences

This project focuses on addressing these issues by building and evaluating an NER system and integrating it into a practical web-based application.

# Objective

The primary objectives of this project are:

- To design a system capable of identifying entities such as Person (PER), Location (LOC), and Organization (ORG)
- To experiment with low-resource datasets such as WikiAnn
- To evaluate the performance of trained models
- To integrate the trained model into a Django-based backend system
- To provide real-time predictions through an interactive interface

The project aims to combine machine learning with practical deployment.

# Project Overview

This project implements an end-to-end Named Entity Recognition system combining machine learning and web development.

- Model training was performed in Google Colab using transformer-based architectures
- Backend was developed using Django REST Framework
- The system accepts user input text and returns predicted entities
- Output is displayed in real-time, making the system interactive

This demonstrates a complete pipeline from data processing to deployment.

# Dataset (WikiAnn)

The WikiAnn dataset is a multilingual dataset designed for Named Entity Recognition tasks.

- Contains annotations for entities such as PER, LOC, and ORG
- Supports multiple languages including low-resource languages
- Used for training and evaluating the NER model

However, the dataset contains noisy annotations, which impacts model performance.

# Model Training and Experimentation

- Initially trained a custom NER model using WikiAnn dataset in Google Colab
- Used HuggingFace Transformers for training
- Observed that due to noisy and limited data, model performance was not optimal
- To improve accuracy, a pretrained model (bert-base-NER) was used for deployment

**Conclusion:** Pretrained models perform better for low-resource datasets.

# Model Training (Colab)

The model was trained using the HuggingFace Transformers library in Google Colab.

- AutoTokenizer was used to convert text into tokens
- AutoModelForTokenClassification was used for NER
- Training was performed on labeled dataset
- Model and tokenizer were saved after training

Due to dataset limitations, a pretrained model was later used to improve performance.

The system follows a simple pipeline for processing input text:

- User inputs a sentence through the interface
- Django backend receives the request via API
- The transformer model processes the text
- Each token is assigned an entity label
- Results are returned in JSON format and displayed

This enables real-time prediction and smooth interaction.

# Technology Stack

The system is built using the following technologies:

- Python for overall development
- HuggingFace Transformers for model implementation
- Django and Django REST Framework for backend
- HTML and JavaScript for frontend
- Google Colab for model training

These tools enable building a complete machine learning application.

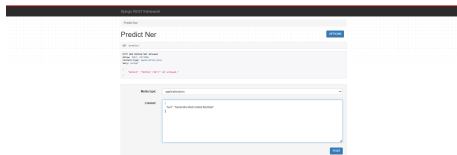
# Key Features

- Real-time Named Entity Recognition
- Detection of Person, Location, and Organization entities
- API-based architecture for easy integration
- Interactive user interface for input and output

The system demonstrates both machine learning capability and practical usability.

# NER Prediction Demo – Input

## Input to the System



Predict Ner

Input: The cat sat on the mat.

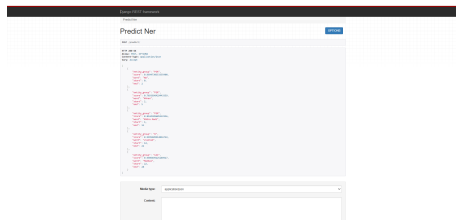
Predict

Model: ner\_model

Predict

**Figure: Input sentence provided through Django REST API interface**

## Output Generated by Model



**Figure: Named entities (Person and Location) identified by the transformer-based NER model**

# Output Example

- Input: Narendra Modi visited Mumbai
- Output:
  - Narendra Modi is identified as a Person (PER)
  - Mumbai is identified as a Location (LOC)

# Results and Observations

- The system successfully detects named entities from input text
- Real-time predictions are achieved through API integration
- Pretrained models provide significantly better performance

This highlights the importance of high-quality training data.

# Limitations

- WikiAnn dataset contains noisy and incomplete annotations
- Custom trained model has limited accuracy
- System struggles with complex or long sentences
- Currently deployed only locally, not on cloud

# Future Work

- Improve dataset quality and annotation
- Extend support to more languages
- Deploy system on cloud platforms
- Enhance user interface and visualization

# Conclusion

- Developed a complete Named Entity Recognition system
- Integrated machine learning model with Django backend
- Achieved real-time entity extraction

The project demonstrates both theoretical understanding and practical implementation of NLP systems.

# Thank You